3D Matrices: Transformations

Questions 1–8

Question 1. Each of the following are an eigenvalue with its corresponding eigenvector for a matrix $\mathbf{M}$
In each case, describe the effect of the transformation in the direction of the given eigenvector:

a) $\lambda = 2, \mathbf{v} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$

b) $\lambda = 4, \mathbf{v} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$

c) $\lambda = 1, \mathbf{v} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$

d) $\lambda = -1, \mathbf{v} = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$

e) $\lambda = -2, \mathbf{v} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

Question 2. A transformation T is represented by the matrix $\mathbf{A} = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

a) Find the eigenvalues of matrix $\mathbf{A}$ and their corresponding eigenvectors.
b) Find $\det \mathbf{A}$ and give a geometrical interpretation of this value.
c) Describe fully the transformation represented by $\mathbf{A}$

Question 3. A transformation T is represented by the matrix $\mathbf{A} = \begin{pmatrix} -1 & 2 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix}$

a) Find the eigenvalues of matrix $\mathbf{A}$ and their corresponding eigenvectors.
b) Find $\det \mathbf{A}$ and give a geometrical interpretation of this value.
c) Describe fully the transformation represented by $\mathbf{A}$

Question 4. A transformation T is represented by the matrix $\mathbf{A} = \begin{pmatrix} 3 & -4 & -2 \\ 0 & -1 & 0 \\ 0 & 2 & 1 \end{pmatrix}$

a) Find the eigenvalues of matrix $\mathbf{A}$ and their corresponding eigenvectors.
b) Find $\det \mathbf{A}$ and give a geometrical interpretation of this value.
c) Describe fully the transformation represented by $\mathbf{A}$

Question 5. Under a linear transformation, T, the point $(3, 4, -1)$ is mapped to the point $(-15, -20, 5)$
For the matrix representing T:

a) Write down the eigenvector.

b) Write down the corresponding eigenvalue.

Question 6. A transformation T is represented by the matrix $\mathbf{A}$. Under the transformation, points on the line with direction $\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$ are stretched by scale factor 2. Points along the x-axis are invariant. Points along the z-axis are stretched by scale factor 3 and reflected across the origin.

a) Describe fully the transformation represented by $\mathbf{A}^n$ giving your answers in terms of n

b) Determine the matrix $\mathbf{A}$

c) Determine the matrix $\mathbf{A}^n$

Question 7. A transformation T is represented by the matrix $\mathbf{A} = \begin{pmatrix} p & q & 0 \\ q & p & 0 \\ 0 & 0 & r \end{pmatrix}$ for non-zero constants p, q and r where $p < q < r$

a) Find the eigenvalues and corresponding eigenvectors of matrix $\mathbf{A}$ giving your answer in terms of p, q and r

b) Show that the three invariant lines are mutually perpendicular.

c) Describe fully the transformation represented by $\mathbf{A}$

Question 8. A transformation T is represented by the matrix

$$\mathbf{A} = \begin{pmatrix} 1-k & k & k \\ -2 & k+1 & 2 \\ 2-2k & k & 2k-1 \end{pmatrix}$$

where k is a constant where $k > 2$

The eigenvalues of $\mathbf{A}$ are $\lambda_1 < \lambda_2 < \lambda_3$

a) Find the eigenvalues of $\mathbf{A}$

The eigenvectors of $\mathbf{A}$ are given by

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \text{ and } \mathbf{v}_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

b) Describe fully the transformation represented by $\mathbf{A}^3$ giving your answers in terms of k